

# Raphael Memmesheimer, Dr. rer. nat.

 <https://raphaelmemmesheimer.github.io>

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## Work Experience

- **Autonomous Intelligent Systems, University of Bonn** 2022 –  
Postdoctoral Researcher
- **Active Vision Working Group, University of Koblenz-Landau** 2016 – 2022  
Research Associate
- **Active Vision Working Group, University of Koblenz-Landau** 2012 – 2016  
Research Assistant
- **Clasen Datensysteme GmbH** 2006 – 2012  
Software Engineer

## Education

- **Dr. rer. nat. (Ph.D.), University of Koblenz-Landau** Computer Science 2016 – 2022  
magna cum laude  
Thesis title: *On the recognition of human activities and the evaluation of its imitation by robotic systems*  
Mentors: Prof. Dr. Dietrich Paulus.
- **M.Sc. Computational Visualistics, University of Koblenz-Landau** 2015 – 2016  
Grade: 1.4, high distinction  
Mentors: Prof. Dr. Dietrich Paulus, Dipl. Inform. Nicolai Wojke.
- **B.Sc. Computational Visualistics** 2010 – 2015  
Grade: 2.2, good  
Mentors: Prof. Dr. Dietrich Paulus, Dipl. Inform. Viktor Seib.

## Research Stays

- **Intelligent Robots and Systems group (IRSg), Instituto Superior Técnico (IST)** 07/2018 – 10/2018  
*Lisbon, Portugal.*

## Research Publications

### Dissertation

- 1 **R. Memmesheimer**, “On the recognition of human activities and the evaluation of its imitation by robotic systems,” Ph.D. dissertation, University of Koblenz and Landau, Germany, 2023.  URL: <https://kola.opus.hbz-nrw.de/frontdoor/index/index/docId/2396>.

### Journal Articles

- 1 E. Kruzhkov, **R. Memmesheimer**, and S. Behnke, “OMCL: Open-vocabulary monte carlo localization,” *IEEE Robotics and Automation Letters (RAL)*, 2026.  DOI: 10.1109/lra.2026.3653333.
- 2 **R. Memmesheimer**, M. Overbeck, D. Beyer, B. Kral, S. Bellmann, S. Schneider, J. Zimmermann, A.-M. Meer, M. Klicic, S. Roth, C. Straßmann, A. Arntz, M. Wessels, J. Kraus, P. Schweidler, T. Schnell, C. Zimmermann, B. Pulver, W. Stork, M. Gersch, S. Behnke, and A. Roennau, “Benchmarking robots for everyday environments: From lab experiments to real-world operations,” *Frontiers in Robotics and AI - Field Robotics (Accepted for publication)*, 2026.

- 3 C. Lenz, M. Schwarz, A. Rochow, B. Pätzold, **R. Memmesheimer**, M. Schreiber, and S. Behnke, "NimbRo wins ANA avatar XPRIZE immersive telepresence competition: Human-centric evaluation and lessons learned," *Int. J. Soc. Robotics*, vol. 17, no. 3, pp. 337–361, 2025. [DOI: 10.1007/S12369-023-01050-9](#).
- 4 M. A. Roa, M. R. Dogar, J. Pagès, C. Vivas, A. Morales, N. Correll, M. Görner, J. Rosell, S. Foix, **R. Memmesheimer**, and F. Ferro, "Mobile manipulation hackathon: Moving into real world applications," *IEEE Robotics Autom. Mag.*, vol. 28, no. 2, pp. 112–124, 2021, ISSN: 1070-9932. [DOI: 10.1109/MRA.2021.3061951](#).

## Conference Proceedings

- 1 J. Bode, **R. Memmesheimer**, and S. Behnke, "El3dd: Extended latent 3d diffusion for language conditioned multitask manipulation," in *Accepted for European Robotics Forum (ERF), Stavanger, Norway, March 2026, 2026*.
- 2 S. Bultmann, **R. Memmesheimer**, J. Nogga, J. Hau, and S. Behnke, "Anticipating human behavior for safe navigation and efficient collaborative manipulation with mobile service robots," in *2026 IEEE/SICE International Symposium on System Integration, SII*, IEEE, 2026, pp. 1071–1078. [DOI: 10.1109/SII64115.2026.11404618](#).
- 3 **R. Memmesheimer**, D. Pavlichenko, E. Kruzhkov, J. Bode, F. Schilke, V. Tutevych, C. Lenz, M. Schreiber, and S. Behnke, "Adaptive domestic service robotics through foundation models for perception, interaction, and action," in *Proceedings of the 2nd German Robotics Conference (GRC) (Accepted for presentation)*, Cologne, Germany, Mar. 2026.
- 4 V. Tutevych\*, **R. Memmesheimer\***, L. Eichler, D. Pavlichenko, F. Schilke, R. Krudewig, and S. Behnke, "Efficient image annotation via semi-supervised object segmentation with label propagation," in *RoboCup 2026: Robot World Cup (Accepted for publication)*, 2026.
- 5 S. Bultmann, **R. Memmesheimer**, J. Nogga, J. Hau, and S. Behnke, "Anticipating human behavior for safe and efficient collaborative mobile manipulation," in *Proceedings of the 1st German Robotics Conference (GRC) (Accepted for presentation)*, Nuremberg, Germany, Mar. 2025.
- 6 V. Schönbach, M. Wiedemann, **R. Memmesheimer**, M. Mosbach, and S. Behnke, "Integration of the tiago robot into isaac sim with mecanum drive modeling and learned s-curve velocity profiles," in *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, IEEE, 2025. [DOI: 10.1109/case58245.2025.11163930](#).
- 7 S. Seliunina, A. Otepleko, **R. Memmesheimer**, and S. Behnke, "Person segmentation and action classification for multi-channel hemisphere field of view lidar sensors," in *IEEE/SICE International Symposium on System Integration, SII 2025, Munich, Germany, January 21-24, 2025*, IEEE, 2025, pp. 817–822. [DOI: 10.1109/SII59315.2025.10871131](#).
- 8 Y. Wang, **R. Memmesheimer**, and S. Behnke, "LIAM: multimodal transformer for language instructions, images, actions and semantic maps," in *19th International Conference on Intelligent Autonomous Systems (IAS) (Accepted for publication)*, 2025. [DOI: 10.1007/978-3-032-33871-6\\_9](#).
- 9 J. Bode, B. Pätzold, **R. Memmesheimer**, and S. Behnke, "A comparison of prompt engineering techniques for task planning and execution in service robotics," in *23rd IEEE-RAS International Conference on Humanoid Robots, Humanoids 2024, Nancy, France, November 22-24, 2024*, IEEE, 2024, pp. 309–314. [DOI: 10.1109/HUMANOIDS58906.2024.10769825](#).
- 10 **R. Memmesheimer**, C. Lenz, M. Schwarz, M. Schreiber, and S. Behnke, "Self-centering 3-dof feet controller for hands-free locomotion control in telepresence and virtual reality," in *2024 IEEE Conference on Telepresence*, IEEE, 2024, pp. 57–60. [DOI: 10.1109/Telepresence63209.2024.10841525](#).
- 11 **R. Memmesheimer**, J. Nogga, B. Pätzold, E. Kruzhkov, S. Bultmann, M. Schreiber, J. Bode, B. Karacora, J. Park, A. Savinykh, and S. Behnke, "Robocup@home 2024 OPL winner NimbRo: Anthropomorphic service robots using foundation models for perception and planning," in *RoboCup 2024: Robot World Cup XXVII*, E. Barros, J. P. Hanna, H. Okada, and E. Torta, Eds., ser. Lecture Notes in Computer Science, vol. 15570, Springer, 2024, pp. 515–527. [DOI: 10.1007/978-3-031-85859-8\\_44](#).
- 12 **R. Memmesheimer**, M. Overbeck, B. Kral, L. Steffen, S. Behnke, M. Gersch, and A. Roennau, "Cleaning robots in public spaces: A survey and proposal for benchmarking based on stakeholders interviews," in *RoboCup 2024: Robot World Cup XXVII*, E. Barros, J. P. Hanna, H. Okada, and E. Torta, Eds., ser. Lecture Notes in Computer Science, vol. 15570, Springer, 2024, pp. 373–385. [DOI: 10.1007/978-3-031-85859-8\\_32](#).

- 13 S. Bultmann, **R. Memmesheimer**, and S. Behnke, "External camera-based mobile robot pose estimation for collaborative perception with smart edge sensors," in *IEEE International Conference on Robotics and Automation, ICRA 2023, London, UK, May 29 - June 2, 2023*, IEEE, 2023, pp. 8194–8200. [DOI: 10.1109/ICRA48891.2023.10160892](#).
- 14 I. Germann, **R. Memmesheimer**, and D. Paulus, "Reduced skeleton representation for action recognition on graph convolutional neural networks," in *IEEE/SICE International Symposium on System Integration, SII 2023, Atlanta, GA, USA, January 17-20, 2023*, IEEE, 2023, pp. 1–5. [DOI: 10.1109/SII55687.2023.10039092](#).
- 15 I. Kramer, **R. Memmesheimer**, and D. Paulus, "Classification of pathological and healthy individuals for computer-aided physical rehabilitation," in *IEEE/SICE International Symposium on System Integration, SII 2023, Atlanta, GA, USA, January 17-20, 2023*, IEEE, 2023, pp. 1–6. [DOI: 10.1109/SII55687.2023.10039185](#).
- 16 B. Pätzold, A. Rochow, M. Schreiber, **R. Memmesheimer**, C. Lenz, M. Schwarz, and S. Behnke, "Audio-based roughness sensing and tactile feedback for haptic perception in telepresence," in *IEEE International Conference on Systems, Man, and Cybernetics, SMC 2023, Honolulu, Oahu, HI, USA, October 1-4, 2023*, IEEE, 2023, pp. 1387–1392. [DOI: 10.1109/SMC53992.2023.10394062](#).
- 17 D. Pavlichenko, G. Ficht, A. Amini, M. Hosseini, **R. Memmesheimer**, A. Villar-Corrales, S. M. Schulz, M. Missura, M. Bennewitz, and S. Behnke, "RoboCup 2022 AdultSize Winner NimbRo: Upgraded perception, capture steps gait and phase-based in-walk kicks," in *RoboCup 2022: Robot World Cup XXV [Bangkok, Thailand, July 11-17, 2022]*, A. Eguchi, N. Lau, M. Paetzel-Prüsmann, and T. Wanichanon, Eds., ser. Lecture Notes in Computer Science, vol. 13561, Springer, 2023, pp. 240–252. [DOI: 10.1007/978-3-031-28469-4\\_20](#).
- 18 M. Schwarz, C. Lenz, **R. Memmesheimer**, B. Pätzold, A. Rochow, M. Schreiber, and S. Behnke, "Robust immersive telepresence and mobile telemanipulation: NimbRo wins ANA avatar XPRIZE finals," in *22nd IEEE-RAS International Conference on Humanoid Robots, Humanoids 2023, Austin, TX, USA, December 12-14, 2023*, IEEE, 2023, pp. 1–8. [DOI: 10.1109/HUMANOIDS57100.2023.10375179](#).
- 19 A. von Gladiss, I. Kramer, N. Theisen, **R. Memmesheimer**, A. C. Bakenecker, T. M. Buzug, and D. Paulus, "Data augmentation for training a neural network for image reconstruction in mpi," in *International Workshop on Magnetic Particle Imaging*, 2022. [DOI: 10.18416/IJMPI.2022.2203058](#).
- 20 **R. Memmesheimer**, S. Häring, N. Theisen, and D. Paulus, "Skeleton-dml: Deep metric learning for skeleton-based one-shot action recognition," in *IEEE/CVF Winter Conference on Applications of Computer Vision, WACV 2022, Waikoloa, HI, USA, January 3-8, 2022*, IEEE, 2022, pp. 837–845, ISBN: 978-1-6654-0915-5. [DOI: 10.1109/WACV51458.2022.00091](#).
- 21 M. Duhme, **R. Memmesheimer**, and D. Paulus, "Fusion-gcn: Multimodal action recognition using graph convolutional networks," in *Pattern Recognition - 43rd DAGM German Conference, DAGM GCPR 2021, Bonn, Germany, September 28 - October 1, 2021, Proceedings*, C. Bauckhage, J. Gall, and A. G. Schwing, Eds., ser. Lecture Notes in Computer Science, vol. 13024, Springer, 2021, pp. 265–281, ISBN: 978-3-030-92659-5. [DOI: 10.1007/978-3-030-92659-5\\_17](#).
- 22 S. Häring, **R. Memmesheimer**, and D. Paulus, "Action segmentation on representations of skeleton sequences using transformer networks," in *2021 IEEE International Conference on Image Processing, ICIP 2021, Anchorage, AK, USA, September 19-22, 2021*, IEEE, 2021, pp. 3053–3057, ISBN: 978-1-6654-4115-5. [DOI: 10.1109/ICIP42928.2021.9506687](#).
- 23 I. Kramer, **R. Memmesheimer**, and D. Paulus, "Customer interaction of a future convenience store with a mobile manipulation service robot," in *2021 IEEE International Conference on Omni-Layer Intelligent Systems, COINS 2021, Barcelona, Spain, August 23-25, 2021*, IEEE, 2021, pp. 1–7, ISBN: 978-1-6654-3157-6. [DOI: 10.1109/COINS51742.2021.9524229](#).
- 24 **R. Memmesheimer**, I. Kramer, V. Seib, N. Theisen, and D. Paulus, "Robotic imitation by markerless visual observation and semantic associations," in *2020 IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC 2020, Ponta Delgada, Portugal, April 15-17, 2020*, IEEE, 2020, pp. 275–280, ISBN: 978-1-7281-7079-4. [DOI: 10.1109/ICARSC49921.2020.9096123](#).
- 25 **R. Memmesheimer**, N. Theisen, and D. Paulus, "Gimme signals: Discriminative signal encoding for multimodal activity recognition," in *IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2020, Las Vegas, NV, USA, October 24, 2020 - January 24, 2021*, IEEE, 2020, pp. 10 394–10 401, ISBN: 978-1-7281-6213-3. [DOI: 10.1109/IROS45743.2020.9341699](#).

- 26 **R. Memmesheimer**, N. Theisen, and D. Paulus, "SL-DML: signal level deep metric learning for multimodal one-shot action recognition," in *25th International Conference on Pattern Recognition, ICPR 2020, Virtual Event / Milan, Italy, January 10-15, 2021*, IEEE, 2020, pp. 4573–4580. [DOI: 10.1109/ICPR48806.2021.9413336](#).
- 27 I. Kramer, N. Schmidt, **R. Memmesheimer**, and D. Paulus, "Evaluation of physical therapy through analysis of depth images," in *28th IEEE International Conference on Robot and Human Interactive Communication, RO-MAN 2019, New Delhi, India, October 14-18, 2019*, IEEE, 2019, pp. 1–6, ISBN: 1944-9445. [DOI: 10.1109/RO-MAN46459.2019.8956435](#).
- 28 **R. Memmesheimer**, I. Kramer, V. Seib, and D. Paulus, "Simitate: A hybrid imitation learning benchmark," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2019, Macau, SAR, China, November 3-8, 2019*, IEEE, 2019, pp. 5243–5249, ISBN: 978-1-7281-4004-9. [DOI: 10.1109/IROS40897.2019.8968029](#).
- 29 **R. Memmesheimer**, I. Kuhlmann, M. Mints, P. Schmidt, C. Korbach, I. Germann, and D. Paulus, "Scratchy: A lightweight modular autonomous robot for robotic competitions," in *2019 IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC 2019, Porto, Portugal, April 24-26, 2019*, L. Almeida, L. P. Reis, and A. P. Moreira, Eds., IEEE, 2019, pp. 1–6, ISBN: 978-1-7281-3558-8. [DOI: 10.1109/ICARSC.2019.8733655](#).
- 30 **R. Memmesheimer**, V. Seib, T. Evers, D. Müller, and D. Paulus, "Adaptive learning methods for autonomous mobile manipulation in robocup@home," in *RoboCup 2019: Robot World Cup XXIII [Sydney, NSW, Australia, July 8, 2019]*, S. K. Chalup, T. Niemüller, J. Suthakorn, and M. Williams, Eds., ser. Lecture Notes in Computer Science, vol. 11531, Springer, 2019, pp. 565–577, ISBN: 978-3-030-35699-6. [DOI: 10.1007/978-3-030-35699-6\\_46](#).
- 31 P. Schneider, **R. Memmesheimer**, I. Kramer, and D. Paulus, "Gesture recognition in RGB videos using human body keypoints and dynamic time warping," in *RoboCup 2019: Robot World Cup XXIII [Sydney, NSW, Australia, July 8, 2019]*, S. K. Chalup, T. Niemüller, J. Suthakorn, and M. Williams, Eds., ser. Lecture Notes in Computer Science, vol. 11531, Springer, 2019, pp. 281–293, ISBN: 978-3-030-35699-6. [DOI: 10.1007/978-3-030-35699-6\\_22](#).
- 32 M. Matamoros, V. Seib, **R. Memmesheimer**, and D. Paulus, "Robocup@home: Summarizing achievements in over eleven years of competition," in *2018 IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC 2018, Torres Vedras, Portugal, April 25-27, 2018*, H. Costelha, J. M. F. Calado, L. C. Bento, N. Lopes, and P. Oliveira, Eds., IEEE, 2018, pp. 186–191. [DOI: 10.1109/ICARSC.2018.8374181](#).
- 33 **R. Memmesheimer**, I. Mykhalchyshyna, and D. Paulus, "Gesture recognition on human pose features of single images," in *9th IEEE International Conference on Intelligent Systems, IS 2018, Funchal, Madeira, Portugal, September 25-27, 2018*, R. Jardim-Gonçalves, J. P. Mendonça, V. Jotsov, M. Marques, J. Martins, and R. E. Bierwolf, Eds., IEEE, 2018, pp. 813–819, ISBN: 1541-1672. [DOI: 10.1109/IS.2018.8710515](#).
- 34 **R. Memmesheimer**, I. Mykhalchyshyna, V. Seib, T. Evers, and D. Paulus, "Homer@unikoblenz: Winning team of the robocup@home open platform league 2018," in *RoboCup 2018: Robot World Cup XXII [Montreal, QC, Canada, June 18-22, 2018]*, D. Holz, K. Genter, M. Saad, and O. von Stryk, Eds., ser. Lecture Notes in Computer Science, vol. 11374, Springer, 2018, pp. 512–523, ISBN: 978-3-030-27543-3. [DOI: 10.1007/978-3-030-27544-0\\_42](#).
- 35 **R. Memmesheimer**, V. Seib, and D. Paulus, "Homer@unikoblenz: Winning team of the robocup@home open platform league 2017," in *RoboCup 2017: Robot World Cup XXI [Nagoya, Japan, July 27-31, 2017]*, H. Akiyama, O. Obst, C. Sammut, and F. Tonidandel, Eds., ser. Lecture Notes in Computer Science, vol. 11175, Springer, 2017, pp. 509–520, ISBN: 978-3-030-00308-1. [DOI: 10.1007/978-3-030-00308-1\\_42](#).
- 36 A. Vanzo, L. Iocchi, D. Nardi, **R. Memmesheimer**, D. Paulus, I. Ivanovska, and G. K. Kraetzschmar, "Benchmarking speech understanding in service robotics," in *Proceedings of the 4th Italian Workshop on Artificial Intelligence and Robotics A workshop of the XVI International Conference of the Italian Association for Artificial Intelligence (AI\*IA 2017)*, Bari, Italy, November 14-15, 2017, S. M. Anzalone, A. Farinelli, A. Finzi, and F. Mastrogiovanni, Eds., ser. CEUR Workshop Proceedings, vol. 2054, CEUR-WS.org, 2017, pp. 34–40. [URL: https://ceur-ws.org/Vol-2054/paper6.pdf](#).
- 37 N. Wojke, **R. Memmesheimer**, and D. Paulus, "Joint operator detection and tracking for person following from mobile platforms," in *20th International Conference on Information Fusion, FUSION 2017, Xi'an, China, July 10-13, 2017*, IEEE, 2017, pp. 1–8. [DOI: 10.23919/ICIF.2017.8009746](#).
- 38 V. Seib, S. Manthe, **R. Memmesheimer**, F. Polster, and D. Paulus, "Team homer@unikoblenz - approaches and contributions to the robocup@home competition," in *RoboCup 2015: Robot World Cup XIX [papers from the 19th*



*Annual RoboCup International Symposium, Hefei, China, July 23, 2015*], L. Almeida, J. Ji, G. Steinbauer, and S. Luke, Eds., ser. Lecture Notes in Computer Science, vol. 9513, Springer, 2015, pp. 83–94. [DOI: 10.1007/978-3-319-29339-4\\_7](#).

- 39 V. Seib, **R. Memmesheimer**, and D. Paulus, “Ensemble classifier for joint object instance and category recognition on RGB-D data,” in *2015 IEEE International Conference on Image Processing, ICIP 2015, Quebec City, QC, Canada, September 27–30, 2015*, IEEE, 2015, pp. 143–147. [DOI: 10.1109/ICIP.2015.7350776](#).

## Books and Chapters

- 1 A. von Gladiss, **R. Memmesheimer**, N. Theisen, A. C. Bakenecker, T. M. Buzug, and D. Paulus, “Reconstruction of 1d images with a neural network for magnetic particle imaging,” in *Bildverarbeitung für die Medizin 2022 - Proceedings, German Workshop on Medical Image Computing, Heidelberg, June 26–28, 2022*, ser. Informatik Aktuell, K. H. Maier-Hein, T. M. Deserno, H. Handels, A. Maier, C. Palm, and T. Tolxdorff, Eds., Springer, 2022, pp. 247–252. [DOI: 10.1007/978-3-658-36932-3\\_52](#).
- 2 V. Seib, **R. Memmesheimer**, and D. Paulus, “A ROS-based system for an autonomous service robot,” in *Robot Operating System (ROS): The Complete Reference (Volume 1)*, ser. Studies in Computational Intelligence, A. Koubaa, Ed., vol. 625, Heidelberg: Springer, Mar. 2016, pp. 215–252, ISBN: 978-3-319-26054-9. [DOI: 10.1007/978-3-319-26054-9\\_9](#).

## Preprints

- 1 C. Korbach, M. D. Solbach, **R. Memmesheimer**, D. Paulus, and J. K. Tsotsos, “Next-best-view estimation based on deep reinforcement learning for active object classification,” *arXiv preprint arXiv:2110.06766*, 2021.

## Project Reports

- 1 **R. Memmesheimer** and S. Behnke, “Learn2Grasp: Learning Human-Like Interactive Grasping Based on Visual and Haptic Feedback. Sachbericht zum Verwendungsnachweis; Förderprogramm: Deutsch-französische Projekte zum Thema Künstliche Intelligenz,” Hannover: Technische Informationsbibliothek, Tech. Rep., Jul. 2026. [DOI: 10.34657/40575](#).
- 2 **R. Memmesheimer** and S. Behnke, “RimA: Roboter im Alltag - Benchmarking von Assistenzrobotern. Förderlinie Roboter für Assistenzfunktionen: Interaktion in der Praxis (RA3) - Transferzentrum RimA - Roboter im Alltag. Sachbericht zum Verwendungsnachweis,” Hannover: Technische Informationsbibliothek, Tech. Rep., May 2026. [DOI: 10.34657/36380](#).

## Awards

### Robotic Competitions

- 🥇 1st 🥈 2nd 🥉 3rd 🇩🇪 National 🇪🇺 European 🌐 International
- Placed 3rd in @Home Open Platform League, Best Poster Award, Best in Class General Purpose Service Robot RoboCup World Cup, Incheon, South Korea. 🥉 🌐 2026
  - Placed 1st in @Home League RoboCup German Open, Cologne. 🥇 🇩🇪
  - Placed 2nd in @Home Open Platform League, Best Poster Award RoboCup World Cup, Salvador, Brazil. 🥈 🌐 2025
  - Placed 1st in @Home League RoboCup German Open, Nuremberg. 🥇 🇩🇪
  - Placed 1st in @Home Open Platform League (World Champion), Waitress Captain (Best in Restaurant Test) RoboCup World Cup, Eindhoven, Netherlands. 🥇 🌐 2024
  - Placed 1st in @Home League RoboCup German Open, Kassel. 🥇 🇩🇪
  - 1st Place (NimbRo) ANA Avatar XPRIZE Challenge, Las Vegas, USA. 🥇 🌐 2022
  - 1st in AdultSize Soccer Competition / 1st in AdultSize Technical Challenge / 1st in AdultSize Drop-In Challenge / Best Humanoid Award RoboCup World Cup, Bangkok, Thailand. 🥇 🌐
  - Placed 3rd in the Customer Interaction Category of the Future Convenience Store Challenge (private attendance) World Robot Summit, Nagoya, Japan. 🥉 🌐 2021
  - Placed 1st in @Home Open Platform League (World Champion) RoboCup World Cup, Sydney, Australia. 🥇 🌐 2019

## Awards (continued)

|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                             |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| ➤ <b>Best in Professional Service Robots League</b> <i>European Robotics League, Bonn (Germany), Koblenz (Germany).</i>                                                                                                                                                                                 |       |      |
| ➤ <b>Best in Consumer Service Robots League</b> <i>European Robotics League, Lisbon, Portugal.</i>                                                                                                                                                                                                      |       |      |
| ➤ <b>Placed 3rd in @Home League</b> <i>RoboCup German Open, Magdeburg.</i>                                                                                                                                                                                                                              |       |      |
| ➤ <b>Placed 1st in @Home Open Platform League (World Champion) / Best Poster Award Open Platform League</b> <i>RoboCup World Cup, Montreal, Canada.</i>                                                                                                                                                 |       | 2018 |
| ➤ <b>Placed 3rd in the Customer Interaction Category of the Future Convenience Store Challenge</b> <i>World Robot Summit, Tokyo, Japan.</i>                                                                                                                                                             |       |      |
| ➤ <b>Best in TBM1: Getting to know my home / Best in TBM2: Welcoming Visitors / Best in TBM3: Catering for Grannie Annie's Comfort / Best in TBM5: GPSR (some of them shared equally with other teams)</b> <i>European Robotics League, Lisbon (Portugal), Edinburgh (Scotland), Barcelona (Spain).</i> |       |      |
| ➤ <b>Placed 1st in @Home League</b> <i>RoboCup German Open, Magdeburg.</i>                                                                                                                                                                                                                              |       |      |
| ➤ <b>Placed 1st in @Home Open Platform League (World Champion)</b> <i>RoboCup World Cup, Nagoya, Japan.</i>                                                                                                                                                                                             |       | 2017 |
| ➤ <b>Finalist ICRA 2017 DJI RoboMaster Mobile Manipulation Challenge</b> , Singapore, Singapore.                                                                                                                                                                                                        |                                                                                          |      |
| ➤ <b>Placed 1st in @Home League</b> <i>RoboCup German Open, Magdeburg.</i>                                                                                                                                                                                                                              |       |      |
| ➤ <b>Placed 2nd DJI RoboMaster Technical Challenge</b> , Shenzhen, China.                                                                                                                                                                                                                               |       |      |
| ➤ <b>Best in TBM1: Getting to know my home, Best in TBM4: Visit my home (Navigation), Best in TBM5: GPSR</b> <i>European Robotics League, Lisbon (Portugal), Peccioli (Italy).</i>                                                                                                                      |       |      |
| ➤ <b>Placed 2nd in @Home League</b> <i>RoboCup European Open, Eindhoven, Netherlands.</i>                                                                                                                                                                                                               |       | 2016 |
| ➤ <b>Finalist in @Home League</b> <i>RoboCup World Cup, Leipzig, Germany.</i>                                                                                                                                                                                                                           |                                                                                          |      |
| ➤ <b>Placed 1st in @Home League (World Champion) / Placed 1st in Speech Recognition and Audio Detection / Best Looking Robot Award</b> <i>RoboCup World Cup, Hefei, China.</i>                                                                                                                          |       | 2015 |
| ➤ <b>1st in overall ranking (together with team SocRob) / Best Team award</b> <i>RoCKIn, Lisboa, Portugal.</i>                                                                                                                                                                                          |     |      |
| ➤ <b>Best Demonstration in RoCKIn@Home track</b> <i>RoCKIn Camp, Peccioli, Italy.</i>                                                                                                                                                                                                                   |   |      |
| ➤ <b>Placed 2nd in @Home League</b> <i>RoboCup German Open, Magdeburg.</i>                                                                                                                                                                                                                              |   |      |
| ➤ <b>1st in the @Home Track / 2nd in Object Recognition</b> <i>RoCKIn, Toulouse, France.</i>                                                                                                                                                                                                            |   | 2014 |
| ➤ <b>Finalist in the @Home League</b> <i>RoboCup German Open, Magdeburg.</i>                                                                                                                                                                                                                            |                                                                                        |      |
| ➤ <b>Best Final Demonstration in the RoCKIn@Home track</b> <i>RoCKIn Camp, Rome.</i>                                                                                                                                                                                                                    |                                                                                        |      |
| ➤ <b>Finalist in the @Home League</b> <i>RoboCup WorldCup, Eindhoven, Netherlands.</i>                                                                                                                                                                                                                  |                                                                                        | 2013 |
| ➤ <b>Placed 3rd in the @Home League</b> <i>RoboCup German Open, Magdeburg.</i>                                                                                                                                                                                                                          |   |      |

### Other Awards

|                                                                                                                                                                                  |  |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------|
| ➤ <b>Best Paper Award (Efficient Image Annotation via Semi-Supervised Object Segmentation with Label Propagation)</b> <i>RoboCup Symposium, Incheon, South Korea.</i>            |  | 2026 |
| ➤ <b>Finalist for Best Oral Paper Award (Robust Immersive Telepresence and Mobile Telemanipulation: Nimbro wins ANA Avatar XPRIZE Finals)</b> <i>Humanoids, Austin, USA.</i>     |  | 2023 |
| ➤ <b>Limbo (private built robot was awarded)</b> <i>Region56+ Award, Koblenz, Germany.</i>                                                                                       |  | 2021 |
| ➤ <b>Placed 1st in the Heart-Met Gesture Recognition Challenge / Placed 1st in the Heart-Met Action Recognition Challenge (AcRec@UniKoblenz)</b> <i>Metrics Project, Online.</i> |  |      |
| ➤ <b>Lehrpreis Sommersemester 2017   Projekt- und Forschungspraktikum: Robbie</b> <i>Lehrpreis der Hochschuldidaktischen Arbeitsstelle, Koblenz, Germany.</i>                    |  | 2017 |

## Funding

|                                                                                                                                                                                                            |               |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|
| ➤ <b>Project Funding</b> , OV6DP: Open Vocabulary 6D Pose Estimation<br><i>euROBIN Cascade Funding</i><br>Role: Principal Investigator                                                                     | 60.000 EUR    | 2024 |
| ➤ <b>Project Funding</b> , Sondierungsprojekt Stadtreinigung<br><i>BMBF</i><br>Role: Proposal Preparation (Contribution), Implementation                                                                   | 53.468,57 EUR | 2023 |
| ➤ <b>Award Money</b> , ANA Avatar XPRIZE Challenge<br><i>XPRIZE Foundation</i><br>Role: Award Winner (with Max Schwarz, Christian Lenz, Bastian Pätzold, Andre Rochow, Michael Schreiber, and Sven Behnke) | 5.000.000 USD | 2022 |

## Funding (continued)

- **Sponsorship**, Sponsoring of PAL Robotics TIAGo Steel edition platform 75.000 EUR    2017  
*PAL Robotics*  
 Role: Applicant

### Smaller Grants & Awards

- **Award Money**, World Robot Summit Prize Money 3rd Place 2.200 EUR    2022  
*World Robot Summit*  
 Role: Award Winner
- **Travel Support**, World Robot Summit Travel Support 1.200 EUR  
*World Robot Summit*  
 Role: Applicant
- **Award Money**, Region56+ Award 7.500 EUR    2021  
*Region56+*  
 Role: Award Winner
- **Sponsorship**, Zukunftsforum 2.000 EUR    2019  
*Energieversorgung Mittelrhein*
- **Invited Talk**, VDI Jahresmitglieder-Versammlung 300 EUR  
*VDI*
- **Travel Support**, IROS Mobile Manipulation Hackathon Travel Support 400 EUR    2018  
*IROS 2018*  
 Role: Applicant
- **Award Money**, Lehrpreis 2017 500 EUR    2017  
*Hochschuldidaktische Arbeitsstelle*  
 Role: Award Winner

## Talks

- **Dein Freund und Helfer - der Assistenzroboter NimbRo** 2026  
*Gesellschaft für Informatik (GI), B-IT, Bonn, Germany*
- **Benchmarking autonomer Serviceroboter im öffentlichen Raum**  
*FZI Open House, Forschungszentrum Informatik (FZI), Karlsruhe, Germany*
- **Best Practices: NimbRo@Home** 2025  
*Guest Lecture, University of Augsburg, (remote), Germany*
- **Advancing Autonomous Service Robots: From Human Intelligence to Embodied Artificial Intelligence in Domestic Environments (Scientific Keynote)**  
*RimA Summer School 2025, Forschungszentrum Informatik (FZI), Karlsruhe, Germany*
- **NimbRo@Home: Autonomous service robots performing domestic tasks**  
*RIG STAGE - Demo Pitches, Automatica, Munich, Germany*
- **Panel-Member for Round Table "Benchmarking and Challenges"**  
*RIG Panel, Automatica, Munich, Germany*
- **Advancing Autonomous Service Robots: From Human Intelligence to Embodied Artificial Intelligence**  
*Invited Research Talk, University of Technology Nuremberg (UTN), Nuremberg, Germany*
- **Embodied AI: Bridging Human Motion Analysis and Autonomous Robotic Actions**  
*Invited Research Talk, Technical University of Braunschweig*
- **From Multimodal Few-Shot Action Recognition towards Imitation in Embodied AI** 2024  
*Invited Research Talk, Dresden (Online)*
- **Multimodal Few-Shot Action Recognition for Embodied AI**  
*Invited Research Talk, Leipzig University, Leipzig, Germany*
- **Multimodal Few-Shot Action Recognition for Embodied AI**  
*Invited Research Talk, University of Lübeck, Lübeck, Germany*
- **Embodied AI: Towards improved human robot interaction by learning from human motion** 2023  
*Research Workshop: Control of Movement in Biology and Robotics, University Bayreuth, Campus Kulmbach, Germany*
- **On the recognition and imitation of human activities on robotic platforms** 2022  
*Advanced Lecture Series on Artificial Intelligence and Autonomous Systems, Bonn Rhein Sieg, Germany (online)*
- **Servicerobotik - Wettbewerbe und Anwendungen**  
*MN-Seminar, Darmstadt, Germany (online)*

## Talks (continued)

- › **Gesture and activity challenge result presentation of AcRec@UniKoblenz**  
*HEART-MET workshop, Sankt Augustin, Germany (online)*

› **Golem - Faszination Künstliche Intelligenz**  
*Kultursalon Koblenz, Koblenz, Germany*

› **Towards adaptive learning in robot competitions**  
*European Robotics Forum, Malaga, Spain*

› **Roboter und künstliche Intelligenz**  
*Guestlecture: Intelligenz, Denken und Problemlösen / Roboter - KI oder Bauplan für eine Seele, Koblenz, Germany*

› **Künstliche Intelligenz in Haushalt- und Servicerobotik**  
*VDI Mittelrhein Jahresversammlung, Koblenz, Germany*

› **homer@Uni Koblenz**  
*Laboratory for Active and Attentive Vision, York University, Toronto, Canada*

› **Lernen durch Demonstration**  
*Guestlecture: Intelligenz, Denken und Problemlösen, Koblenz, Germany*

› **RoboCup 2017 / Where is HRI and where can it go to in RoboCup? RO-MAN Workshop: HRI for Service Robots in RoboCup@Home**  
*IEEE International Symposium on Robot and Human Interactive Communication, Lisbon, Portugal*

› **Report from ERL Service Robots RO-MAN Workshop: HRI for Service Robots in RoboCup@Home**  
*IEEE International Symposium on Robot and Human Interactive Communication, Lisbon, Portugal*

› **Team's experience in ERL - Service Robots European robotics competitions and challenges: status quo and lessons learned**  
*European Robotics Forum, Edinburgh, Scotland*

› **Navigation and Mapping of Team homer@UniKoblenz**  
*European Robotics League, Peccioli, Italy*

2021

2020

2019

2018

2017

## Academic Activities

- › RoboCup Symposium  
*Session Chair, Task Planning*

› IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)  
*Associate Editor*

› Center for Robotics, University of Bonn  
*Founding Member*

› European Laboratory for Learning and Intelligent Systems (ELLIS) PhD Program  
*Evaluation Committee*

› RoboCup@Home Eindhoven, Netherlands  
*Organizing Committee RoboCup@Home*

› Institute for Computational Visualistics, University of Koblenz-Landau  
*Marketing*

› University of Koblenz-Landau  
*Member of appointments committee, Institute for Computational Visualistics*

› Institute for Computational Visualistics, University of Koblenz-Landau  
*Teaching staff representative for head of institute sessions*

› RoboCup German Open (Magdeburg)  
*League Co-Chair (RoboCup@Home), together with PD Dr.-Ing. Sven Wachsmuth*

› RoboCup@Home Sydney, Australia  
*Organizing Committee RoboCup@Home*

› RoboCup@Home Montreal, Canada  
*Technical Committee RoboCup@Home*

› RoboCup@Home Nagoya, Japan  
*Organizing Committee RoboCup@Home*

› RoboCup@Home Leipzig, Germany  
*Organizing Committee RoboCup@Home*

2026

2025 –

2024 –

2024

2021 – 2022

2019 / 2020

2016 – 2022

2018 – 2020

2019

2018

2017

2016



## Reviewing Activities

- › *Advanced Robotics, AR* 2026
- › *Künstliche Intelligenz / German Journal of Artificial Intelligence, KI*
- › *IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS* 2020 – 2026
- › *IEEE International Conference on Robotics and Automation, ICRA* 2019 – 2026
- › *IEEE International Conference on Robot and Human Interactive Communication, RO-MAN* 2019 – 2024
- › *IEEE Robotics and Automation Letters, RA-L* 2019 – 2026
- › *IEEE Signal Processing Letters, IEEE SPL* 2023
- › *Springer International Journal of Computer Vision, IJCV*
- › *IEEE International Conference on Humanoid Robots, Humanoids* 2023 – 2024
- › *IEEE/SICE International Symposium on System Integrations, SII* 2024
- › *ACM Transactions on Human-Robot Interaction, THRI*
- › *Workshop Toward Robot Avatars, Avatar Workshop* 2023 – 2024

## Teaching Experience

- › **Lecture (University of Bonn)** WS 25/26  
Cognitive Robotics
- › **Lab (University of Bonn)** since SS 22  
Cognitive Robotics
- › **Projektgruppe (University of Bonn)**  
Kognitive Robotik
- › **Seminar (University of Bonn)**  
Cognitive Robotics
- › **Research Internship (University of Koblenz-Landau, Koblenz)** SS 21 – WS 21/22  
Multimodal Action Recognition
- › **Project Internship (University of Koblenz-Landau, Koblenz)**  
Multimodal Action Recognition
- › **Seminar (University of Koblenz-Landau, Koblenz)** WS 20/21 – WS 21/22  
Machine Learning for Computer Vision
- › **Lecture (University of Koblenz-Landau, Koblenz)** WS 18, WS 19, WS 20  
Autonomous Mobile Robots
- › **Seminar (University of Koblenz-Landau, Koblenz)** SS 17, SS 18, SS 19  
RoboCup@Home Accompanying Seminar
- › **Project internship (University of Koblenz-Landau, Koblenz)** WS 16/17 – WS 18/19  
RoboCup@Home
- › **Research internship (University of Koblenz-Landau, Koblenz)**  
RoboCup@Home

## Supervised Theses

- › **Language-Conditioned Multitask Manipulation Using Diffusion Models** 2025  
*Jonas Bode, M.Sc.*
- › **Action Transcript Prediction from Multi-Modal Environments using Vision-Language Models** 2024  
*Yihao Wang, M.Sc.*
- › **Neural Architecture Search for Skeleton-based Action Recognition** 2022  
*Christian Brozmann, M.Sc.*
- › **Sign language recognition using convolutional neural networks**  
*Matthias Wellstein, B.Sc.*
- › **Action Recognition on Egocentric Video Sequences in Natural Environments**  
*Katrina Mahlendorf, B.Sc.*
- › **Action Recognition on Skeleton Sequences Using Graph Convolutional Neural Networks** 2021  
*Ida Germann, B.Sc.*

## Supervised Theses (continued)

|                                                                                                                                             |  |      |
|---------------------------------------------------------------------------------------------------------------------------------------------|--|------|
| › <b>Next-Best-View Estimation based on Deep Reinforcement Learning for Active Object Classification</b><br><i>Christian Korbach, M.Sc.</i> |  | 2020 |
| › <b>Multimodal Action Recognition using Graph Convolutional Neural Networks</b><br><i>Michael Duhme, M.Sc.</i>                             |  |      |
| › <b>Action Segmentation on Representations of Skeleton Sequences using Transformer Networks</b><br><i>Simon Häring, M.Sc.</i>              |  |      |
| › <b>Evaluation verschiedener Netzwerkarchitekturen für Aktionserkennung auf Signalrepräsentationen</b><br><i>Alex Weissörtel, B.Sc.</i>    |  |      |
| › <b>Generative Adversarial Networks for Image Data Augmentation</b><br><i>Robin Bartsch, B.Sc.</i>                                         |  | 2019 |
| › <b>Adaptives Lernen durch Demonstration mittels Objektinteraktion</b><br><i>Tobias Evers, B.Sc.</i>                                       |  |      |
| › <b>Imitating Manipulation Tasks Using Generative Adversarial Networks</b><br><i>Ivanna Kramer, M.Sc.</i>                                  |  |      |
| › <b>Simultaneous Action and Object Classification in Videos using Convolutional Neural Networks</b><br><i>Lukas Debal, M.Sc.</i>           |  |      |
| › <b>Visual Object Tracking and Prediction Using Recurrent Neural Networks</b><br><i>Thies Möhlenhof, M.Sc.</i>                             |  |      |
| › <b>Detection of Human-Object Interactions</b><br><i>Jasvinder Kaur, M.Sc.</i>                                                             |  |      |
| › <b>Reinforcement Learning for Robot Manipulation Tasks</b><br><i>Jannis Eisenmenger, B.Sc.</i>                                            |  |      |
| › <b>Markerless Action Recognition</b><br><i>Gregor Heuer, M.Sc.</i>                                                                        |  | 2018 |
| › <b>Object detection using convolutional neural networks</b><br><i>Simon Häring, B.Sc.</i>                                                 |  |      |
| › <b>Inverse Reinforcement Learning for Robotic Manipulation Tasks</b><br><i>Nick Theisen, M.Sc.</i>                                        |  |      |
| › <b>Simultaneous Object Localization and Classification based on Semantic Segmentation</b><br><i>Lukas Buchhold, B.Sc.</i>                 |  |      |
| › <b>Evaluation von physiotherapeutischen Übungen durch Analyse von Tiefenbildern</b><br><i>Niko Schmidt, B.Sc.</i>                         |  | 2017 |

## Shortlist

|                                                                               |           |      |
|-------------------------------------------------------------------------------|-----------|------|
| › <b>University of Braunschweig (W2), Professorship for AI-based Robotics</b> | 2nd place | 2025 |
| › <b>University of Luebeck (W1)</b>                                           | 2nd place | 2023 |

## Media Coverage

|                                                                                                          |                 |            |
|----------------------------------------------------------------------------------------------------------|-----------------|------------|
| › Radio Bonn/Rhein-Sieg: Team NimbRo der Uni Bonn hat den beste Haushaltsroboter Deutschlands entwickelt | <b>Radio</b>    | 01.04.2026 |
| › Deutschlandfunk: RoboCup German Open 2026 – Weltweit größter Wettbewerb für intelligente Roboter       | <b>Radio</b>    | 11.03.2026 |
| › Bild der Wissenschaft: Die Champions vom Rhein                                                         | <b>Magazine</b> | 14.01.2025 |
| › WDR Lokalzeit: Bonner Haushaltsroboter sind Weltmeister                                                | <b>TV</b>       | 27.09.2024 |
| › IEEE Spectrum: Your robotic avatar is almost ready                                                     | <b>Magazine</b> | 16.04.2023 |
| › RLP.TV                                                                                                 | <b>TV</b>       | 18.07.2019 |
| › RLP.TV                                                                                                 | <b>TV</b>       | 31.05.2019 |
| › ARD Tagesschau (12:00)                                                                                 | <b>TV</b>       | 19.10.2018 |
| › SWR Landeschau: Deutschland ist Roboter-Weltmeister                                                    | <b>TV</b>       | 02.07.2018 |

## Media Coverage (continued)

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|                                                                              |           |            |
|------------------------------------------------------------------------------|-----------|------------|
| › Campus TV - Zu Besuch beim Roboterteam                                     | <b>TV</b> | 19.03.2018 |
| › TV Mittelrhein - TIAGo der Serviceroboter Studiotalk                       | <b>TV</b> | 07.03.2018 |
| › STV: Home-help robots put through paces in tournament                      | <b>TV</b> | 26.01.2018 |
| › SAT.1 - 17:30                                                              | <b>TV</b> | 02.11.2017 |
| › SWR Landeschau, Haushalts-Roboter aus Koblenz gewinnt Weltmeistertitel     | <b>TV</b> | 15.08.2017 |
| › tech360.tv news roundup - Week 1 June 2017                                 | <b>TV</b> | 01.06.2017 |
| › SWR - Kollege Roboter kommt gleich                                         | <b>TV</b> | 03.11.2016 |
| › Campus TV (Tag der Computervisualistik)                                    | <b>TV</b> | 25.07.2016 |
| › SWR Landesschau                                                            | <b>TV</b> | 19.07.2016 |
| › DASDING! (Koblenzer Roboter Lisa brät Spiegeleier)                         | <b>TV</b> | 27.05.2016 |
| › ARD - W wie Wissen (Spiegelei und Kaffee, bitte! Roboter als Servicekraft) | <b>TV</b> | 27.02.2016 |
| › SAT.1 - 17:30 (Roboter Lisa hilft beim Leben)                              | <b>TV</b> | 11.12.2015 |